

Engine Parameters & Physics-Informed Model

Reference specification for the AI-enabled digital twin — parameter definitions, operating ranges, and the expected-behavior model, with sources.

This document grounds the parameters and physics rules used by the digital twin's "expected behavior" layer in published reference data — mainly the operator's manual for the Rotax 912/914 engine family, since a turbocharged 912-series engine (the Rotax 914) is the actual powerplant of the IAI Heron, a MALE UAV operated by India and the closest public reference to the class of engine this problem statement targets. Every numeric range below is cited; anywhere a value is a design choice rather than a sourced fact, it is marked as such.

1. Parameter Reference Table

These are the sensor channels the digital twin ingests. Ranges are for a Rotax 912/914-class four-cylinder aero piston engine — the nearest publicly documented match to the engine class DRDO's problem statement targets — and should be treated as a calibration starting point, re-fitted once real engine specifications are available, exactly as the honest-scoping note in the base plan states.

Parameter	Unit	Typical range	Why it matters	Source
Engine speed (RPM)	rpm	Idle ~1,400–2,100; max continuous 5,500; takeoff 5,800 (5 min max)	Primary driver of nearly every other parameter — heat, pressure, fuel flow, vibration all scale with RPM.	[1][2]
Cylinder head temperature (CHT)	°C	Normal 100–190; max 150 (some variants 235)	Direct indicator of combustion heat and cooling-system health; sustained high CHT accelerates detonation risk and cylinder wear.	[1]
Exhaust gas temperature (EGT)	°C	~450–800 normal; max ~900 (1,652°F)	Reflects mixture (AFR) and combustion quality; abnormal EGT spread across cylinders flags misfire or injector faults.	[2][7]
Oil temperature	°C	Min 50 (warm-up); normal 88–110; caution above ~120	Governs oil viscosity, which sets the expected oil pressure — the single most-watched lubrication health signal.	[1][4]
Oil pressure	psi (bar)	Normal 29–73 (2.0–5.0 bar) above 3,500 rpm; min 12 psi below 3,500 rpm;	Falling pressure at a given RPM/temp is the earliest reliable sign of a developing lubrication fault (leak, pump wear, filter blockage).	[1][2][3]

Parameter	Unit	Typical range	Why it matters	Source
		cold-start max ~102 psi		
Fuel flow	L/h	~13–18.5 at 75% power; ~25 at max continuous; ~27 at takeoff	Should track power setting predictably; deviation flags injector, mixture, or fuel-system faults.	[3]
Brake-specific fuel consumption (BSFC)	g/kWh	~285 at max continuous power	Engine-level efficiency constant used to derive expected fuel flow from estimated power output.	[3]
Vibration (broadband RMS)	mm/s (or g)	Baseline low at cruise RPM; ISO industrial zones A/B ≈7.1, B/C ≈14.0, C/D ≈28.0 mm/s RMS (adapted reference, not engine-specific)	Rising vibration relative to a healthy RPM-matched baseline is a strong early indicator of imbalance, bearing wear, or prop damage.	[8]
Air/fuel ratio (AFR) / lambda	ratio	Stoichiometric 14.7:1 ($\lambda=1$); operating range roughly 12:1 (rich) – 16:1 (lean)	Sets combustion completeness; drives the EGT physics rule and is the key signature for injector/mixture faults.	[7]
Battery / alternator (generator) voltage	V	Resting ~12.0–12.8; charging in flight ~13.5–14.2	Confirms electrical system health, which the ECU/FADEC and sensors themselves depend on — a silent single point of failure if ignored.	[11][12]
Injection / ignition timing	° BTDC	~18–26, ECU/CDI-map dependent on RPM and load	Off-nominal timing changes combustion phasing and shows up as correlated EGT/CHT drift without a corresponding RPM change.	engine ECU/CDI map (OEM-specific, not independently sourced here)
Altitude	m (ft)	0–7,000 m (0–23,000 ft) typical MALE UAV operating band; ceiling up to ~30,000–35,000 ft class	Sets air density, which is the single biggest confound on CHT, oil temp, and power output — must be corrected for before comparing to sea-level baselines.	[9][10]
Ambient (outside air) temperature	°C	–40 to +45 (mission/region dependent)	Second major environmental confound alongside altitude; both feed directly into every expected-value formula below.	[5]

Note on vibration: aviation piston engines don't publish g-force or mm/s fault thresholds the way industrial machinery standards do. The ISO 20816 values above are an industrial reciprocating-machinery framework used here only to justify a zone-based (A/B/C/D) severity structure — the actual thresholds must be learned from your own healthy-baseline data, not read off this standard directly.

2. Physics-Informed Model — Detailed

2.1 Purpose and design philosophy

A physics-informed model exists to answer one question at every timestep: "given the current RPM, altitude, and ambient temperature, what SHOULD each parameter read if the engine is healthy?" The digital twin never compares a live reading against a fixed threshold — it compares it against this computed expectation. This is the exact gap the problem statement identifies in current threshold-based monitoring, and it is also what lets the same model work across different missions and climates without retraining, because the expectation itself moves with the operating condition.

Every rule below follows the same pattern:

```
expected_value = f(RPM, altitude, ambient_temp, [oil_temp | AFR | throttle])
```

The output of every rule feeds into one deviation calculation, which is reused everywhere downstream (health index, anomaly features, RUL trend input):

```
deviation_pct = (actual - expected) / expected × 100
```

2.2 Foundation layer: standard atmosphere

Altitude affects almost every other parameter through air density, so this is computed once and reused by every other rule. The International Standard Atmosphere (ISA) model defines sea-level reference conditions and how temperature, pressure, and density fall off with altitude in the troposphere (0–11 km), which covers the full operating envelope of a MALE UAV.

```
T(h) = 288.15 - 0.0065 × h [Kelvin, h in meters]
ρ(h) / ρ0 = (1 - 2.2558×10-5 × h)4.2559 [ρ0 = 1.225 kg/m3 at sea level]
```

Reference sea-level values: pressure 1,013.25 hPa, temperature 15°C (288.15 K), density 1.225 kg/m³, lapse rate -0.0065 K/m up to the tropopause. [5][6]

Worked example — density ratio at 4,200 m (a typical MALE UAV cruise altitude):

```
σ = (1 - 2.2558×10-5 × 4200)4.2559 = (1 - 0.0948)4.2559 ≈ 0.633
```

At this altitude the engine is breathing about 63% of sea-level air density — every downstream rule that references σ inherits this correction automatically.

2.3 Cylinder head temperature (CHT) rule

CHT rises with combustion heat (RPM-driven) and falls with cooling airflow, which is itself density-dependent. A naive model would only look at RPM and miss the altitude effect entirely — this is the specific failure mode a threshold-only system has.

```
CHT_expected = CHT_base(RPM) + 0.5 × (T_ambient - 15) - k_alt × (σ - 1)
```

CHT_base(RPM) is a curve fit against your own synthetic-healthy dataset, anchored to the manual's stated operating band of roughly 100–190°C rising toward the 150°C limit near redline [1]. k_alt is a calibration constant (starting estimate 15) representing how much CHT rises as air thins — set this by running your synthetic generator at fixed RPM across a range of altitudes and fitting the resulting curve; it is not itself a published Rotax constant.

2.4 Oil temperature and oil pressure — a coupled pair

These two must be modeled together, because oil pressure's expected value depends on oil temperature (through viscosity), not on RPM alone. This coupling is the single most DRDO-relevant physics rule in the whole model — it is the difference between "pressure below X = alarm" (threshold-based, exactly what the problem statement wants replaced) and "pressure below what THIS temperature and RPM predict = alarm" (physics-informed).

```
oil_temp_expected = 55 + 0.009 × RPM + 0.6 × T_ambient - 8 × (σ - 1)
viscosity_factor = exp(-0.02 × (oil_temp_actual - 90))
oil_pressure_expected = 0.013 × RPM × viscosity_factor
```

The oil_temp formula is anchored to the manual's stated normal operating band of 88–110°C (190–230°F) and minimum operating temperature of 50°C [1][4]. The oil_pressure formula is anchored to the manual's stated normal band of 29–73 psi above 3,500 rpm, falling to a 12 psi floor below 3,500 rpm [1][2][3] — note this is a two-piece curve (there is a step change in the minimum-admissible value at 3,500 rpm), so the model should branch on RPM rather than use one continuous formula across the whole range.

Worked example at RPM=4,200, oil_temp_actual=92°C:

```
viscosity_factor = exp(-0.02 × (92 - 90)) = exp(-0.04) ≈ 0.961
oil_pressure_expected = 0.013 × 4200 × 0.961 ≈ 52.5 psi
```

If the live reading is 51 psi, deviation is only -2.9% — normal. If it's 40 psi, deviation is -23.8% at the same RPM and oil temperature — a real fault signature, not just "the oil is hot."

2.5 Exhaust gas temperature (EGT) rule

EGT is primarily a function of RPM (load) and mixture (AFR). This is the rule that ties combustion chemistry into the model — a lean or rich mixture is exactly what an injector or mixture-control fault produces, and it shows up here before it shows up anywhere else.

```
EGT_expected = 500 + 0.06 × RPM + 40 × (AFR_actual - 14.7)
```

The constant 14.7 is the stoichiometric air/fuel ratio for gasoline — the chemically ideal mixture — not a tuned value [7]. Richer mixtures (AFR below 14.7) run cooler; leaner mixtures run hotter, up toward the manual's stated EGT ceiling near 900°C [2]. If you don't have a direct AFR/lambda sensor in your synthetic data, approximate AFR from commanded fuel flow versus RPM-implied airflow, or omit the AFR term and rely on RPM alone for MVP, adding the AFR term in Phase 3/4.

2.6 Fuel flow rule

Fuel flow should track commanded power in a predictable, near-linear way. This rule is the one place you can pull a manufacturer constant directly instead of curve-fitting one — BSFC is a published, engine-level efficiency number.

```
power_estimate_kw = k_power × RPM × throttle_fraction    [k_power fit from your
synthetic data]
fuel_flow_expected_Lph = (BSFC × power_estimate_kw) / fuel_density_g_per_L
```

BSFC = 285 g/kWh at max continuous power, taken directly from the operator's manual [3]. Using a typical gasoline density of ~720 g/L: at max continuous power (58 kW per the manual [2]), expected fuel flow works out to $(285 \times 58) / 720 \approx 23$ L/h — close to, though not identical to, the manual's separately tabulated 25 L/h figure. This 2 L/h gap is a real, published inconsistency in Rotax's own documentation between its power-based and directly-tabulated fuel figures [13] — worth stating explicitly in your pitch rather than hiding it, since it shows you've read the source material critically rather than skimmed it.

2.7 Vibration baseline

Unlike the rules above, vibration has no clean closed-form physics equation at the fidelity available here — it depends on mechanical balance, prop condition, and mount stiffness, none of which are in your sensor list. Build this as an empirical baseline instead of a formula:

- Run your synthetic generator in healthy mode across the full RPM range and record vibration.
- Fit a smooth curve (vibration tends to rise with RPM, with resonance bumps at specific bands from propeller harmonics) — this curve IS your "expected" function for vibration.
- Compare live vibration against this RPM-matched baseline, not a flat number, exactly as with every other parameter.
- Use the ISO 20816 zone structure (A/B/C/D) only as a severity-labeling convention for how far above baseline a reading has drifted, not as the threshold values themselves [8].

2.8 How the parameters combine — one full worked example

Flight condition: RPM 4,200; altitude 4,200 m; ambient temperature 18°C; oil temperature actual 92.3°C; AFR actual 14.9 (slightly lean).

```
σ = 0.633    (from §2.2)
CHT_expected ≈ 150 + 0.5×(18-15) - 15×(0.633-1) ≈ 157.0°C
oil_temp_expected = 55 + 0.009×4200 + 0.6×18 - 8×(0.633-1) ≈ 96.6°C
viscosity_factor = exp(-0.02×(92.3-90)) ≈ 0.955
oil_pressure_expected = 0.013×4200×0.955 ≈ 52.2 psi
EGT_expected = 500 + 0.06×4200 + 40×(14.9-14.7) ≈ 760.0°C
```

Every one of these five numbers is computed from the same four inputs (RPM, altitude, ambient temp, AFR/oil-temp) — this is what "physics-informed" means in practice: one small set of environmental readings drives the expected value of every monitored channel simultaneously, and every channel's actual-vs-expected gap becomes a feature for the health index, the anomaly detector, and the RUL model in the same pass.

2.9 Calibration methodology

The formulas above are structurally correct (the right variables combined the right way) but their constants are starting estimates, not DRDO-verified values, because no public dataset exists for a classified/operational engine. State this plainly in the pitch — it is expected, not a weakness:

- Generate a large healthy-only synthetic dataset spanning the full RPM/altitude/ambient-temperature envelope.
- For each rule, fit the unknown constants (k_{alt} , k_{power} , the base CHT curve, etc.) by regression against that healthy dataset — the physics form is fixed, only the coefficients are learned.
- Validate by checking that fitted expected values track the manual's stated normal operating bands (§1) across the RPM range, not just at one point.
- When real engine telemetry becomes available, re-run the same fitting procedure — the architecture doesn't change, only the constants do. This is the "engine-agnostic, recalibrated per engine type" framing from the base plan's honest-scoping section.

2.10 Limitations to state explicitly to the judges

- All operating ranges are sourced from a Rotax 912/914-class engine (via its use in the IAI Heron MALE UAV [9][10]) — the closest public reference, not DRDO's actual (classified) engine specification.
- The vibration model has no closed-form physics; it is empirical and baseline-relative, using ISO 20816's zone concept only as a severity-labeling convention [8].
- The fuel-flow rule surfaces a known ~2 L/h internal inconsistency in Rotax's own published figures [3][13] — flagged rather than silently resolved.
- Injection/ignition timing has no independently sourced public range here; it is ECU/CDI-map dependent and engine-specific, and would need to come from the actual FADEC/ECU documentation for the target engine.

3. References

- [1] BRP-Rotax, "Operator's Manual for ROTAX Engine Type 912 Series" (OM-912, Edition 4/Rev.0), Section 2.2 Operating Limits (912 S/ULS) — CHT max 150°C, coolant exit temp max 120°C.
https://avsport.org/acft/Rotax/OM_912_Series_ED4_R0.pdf
- [2] Rotax 912 ULS performance data sheet (DCDI) — takeoff 5,800 rpm / max continuous 5,500 rpm, EGT max 1,652°F (900°C), oil pressure max 72 psi / min 22 psi @ 2,800 rpm / cold-start max 102 psi, TBO 1,500 hrs.
https://avsport.org/acft/Rotax/912uls_info.pdf
- [3] ROTAX 912 UL/ULS Operator's Manual, Section 8 (Fuel consumption) and Section 10.1.2 (Oil pressure) — via ManualsLib. Oil pressure 2.0–5.0 bar (29–73 psi) above 3,500 rpm, min 0.8 bar (12 psi) below 3,500 rpm; fuel consumption at takeoff/max continuous/75% = 27/25/18.5 L/h (912 ULS); specific consumption 285 g/kWh. <https://www.manualslib.com/manual/1083170/Rotax-912-UL.html>
- [4] Rotax Owner community, citing ROTAX 912 Operator's Manual §10.1.2.4 — normal operating oil temperature 190–230°F (88–110°C); minimum operating oil temperature 50°C (122°F). <https://www.rotax-owner.com/en/912-914-technical-questions/5070-acceptable-oil-temperatures-how-to-achieve-them>
- [5] SKYbrary Aviation Safety, "International Standard Atmosphere (ISA)" — sea-level pressure 1,013.2 mb, temperature 15°C, density 1.225 kg/m³, lapse rate ≈2°C/1,000 ft.
<https://skybrary.aero/articles/international-standard-atmosphere-isa>
- [6] "Satellite-mediated quantum atmospheric links / InvSim algorithm" (arXiv:2511.03745) — ISA troposphere density formula $\rho = 1.225 \times (1 - 2.2558 \times 10^{-5} h)^{4.2586}$. <https://arxiv.org/pdf/2511.03745>
- [7] x-engineer.org, "Air fuel ratio" — stoichiometric AFR for gasoline ≈14.7:1; combustion range roughly 6:1–20:1. <https://x-engineer.org/air-fuel-ratio/>
- [8] ISO 20816-8:2018, "Mechanical vibration — Measurement and evaluation of machine vibration — Part 8: Reciprocating compressors" — zone boundaries A/B 7.1 mm/s, B/C 14.0 mm/s, C/D 28.0 mm/s RMS, as summarized by Vibromera. <https://vibromera.eu/calculators/vibration-iso20816-8-compressors/>
- [9] Aviation Week, "Program Dossier: IAI Heron" — Heron powered by a single Rotax 914 4-cylinder piston engine, 115 hp takeoff / 100 hp continuous to 15,000 ft. <https://aviationweek.com/defense/program-dossier-iai-heron>
- [10] GlobalSecurity.org, "Heron" — MALE UAV, turbocharged Rotax 914 engine, ceiling 30,000 ft, endurance up to 50 hours, operated by Indian defence forces among others.
<https://www.globalsecurity.org/military/world/israel/heron.htm>
- [11] Rotax 912iS/915iS Electrical Power System documentation — generator nominal output 14.2 V.
<https://superpetrelusa.com/wp-content/uploads/2024/02/ROTAX-912iS-ELECTRICAL-POWER-SYSTEM.pdf>
- [12] wiringpedia.com, "Wiring Diagram For Rotax 912 Voltage Regulator" — regulated DC output typically 13.5–14.2 V. <https://www.wiringpedia.com/wiring-diagram-for-rotax-912-voltage-regulator/>
- [13] rotax-owner.com community thread, "912ULS Fuel Consumption Data" — documents the discrepancy between the Operator's Manual's power-graph-derived and directly-tabulated fuel flow figures.
<https://dev310.rotaxowner.com/en/912-914-technical-questions/11053-912uls-fuel-consumption-data>

All sourced figures above reflect a Rotax 912/914-class general-aviation/UAV engine and are offered as the nearest credible public reference for calibrating this system's physics-informed model — not as DRDO-verified specifications for a specific classified platform.